



More loops

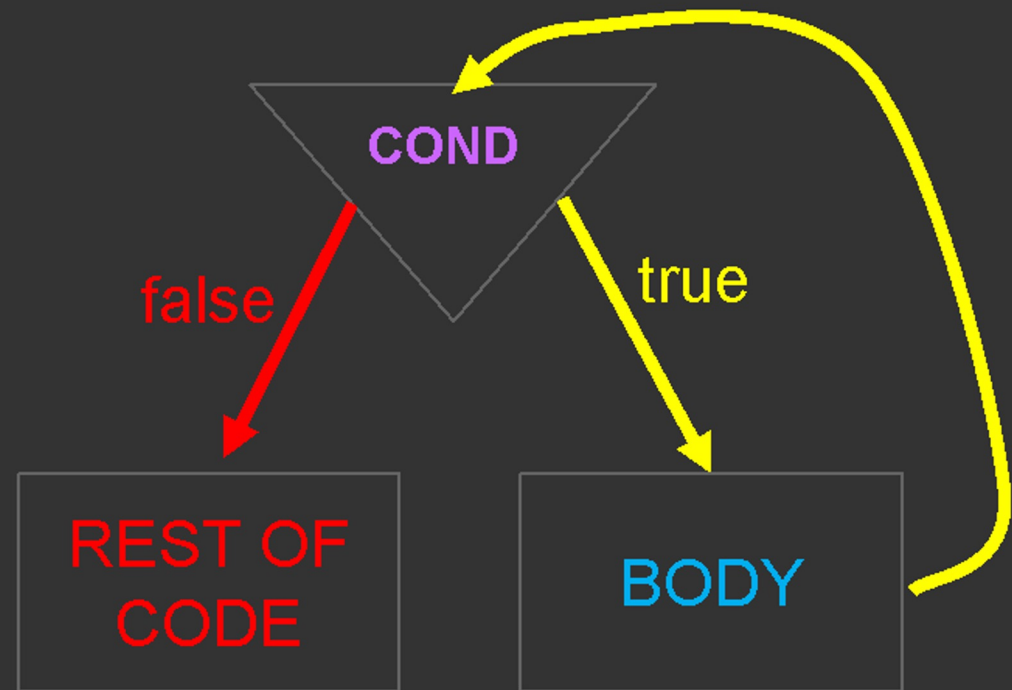
CS 141

(slides by Jaime Spacco; edits by Vera Kazakova and David Bunde)

Administrivia

while loop syntax

```
while (COND) {  
    BODY  
}  
REST OF CODE
```



```
// Let's count to 1000 (how would you code this up?)
```

```
// Pseudocode here for the operation:
```

```
//   Loop 1000 times, using a counter to keep track of the iteration number
```

```
//   print the counter on each iteration
```

//Let's count to 1000

```
int count = 0;  
while (count < 1000) {  
    System.out.print(count + " ");  
    count++;  
}
```

What does this print?

```
int i=5;  
while (i>=0) {  
    System.out.print(i);  
    i++;  
}
```

A: 5 4 3 2 1 0

B: 5 4 3 2 1

C: 5 4 3 2 1 0 -1

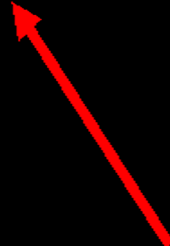
D: the loop never executes

E: it prints forever

What does this print?

```
int i=5;  
while (i>=0) {  
    System.out.print(i);  
    i++;  
}
```

Probably meant to do `i--` here!



A: 5 4 3 2 1 0

B: 5 4 3 2 1

C: 5 4 3 2 1 0 -1

D: the loop never executes

E: it prints forever

How do we kill an infinite loop?

Click in the terminal

Control-C

Let's demo this!

What does this print?

```
int sum = 0;
int i = 0;
while(i<10) {
    sum += i;
    i += 4;
}
System.out.println(sum);
```

A: 4

B: 8

C: 12

D: 16

E: The loop is infinite

What does this print?

```
int sum = 0;
int i = 0;
while(i<10) {
    sum += i;
    i += 4;
}
System.out.println(sum);
```

A: 4

B: 8

C: 12 0+4+8

D: 16

E: The loop is infinite

Let's trace the code:

<u>i</u>	<u>i<10?</u>	<u>sum</u>
0	true	0+0 = 0
0+4 = 4	true	0+4 = 4
4+4 = 8	true	4+8 = 12
8+4 = 12	false	Still 12

What does this print?

```
int sum = 0;
int i = 0;
while(i>100) {
    sum += i;
    i++;
}
System.out.println(sum);
```

- A: 0
- B: 100
- C: $1 + 2 + 3 + \dots + 100$
- D: $1 + 2 + 3 + \dots + 99$
- E: the loop is infinite

What does this print?

```
int sum = 0;
int i = 0;
while(i>100) {
    sum += i;
    i++;
}
System.out.println(sum);
```

A: 0

B: 100

C: 1 + 2 + 3 + ... + 100

D: 1 + 2 + 3 + ... + 99

E: the loop is infinite

i is not greater than 100 on the first iteration,
so the loop body never executes
and the loop ends immediately.

What does this print?

```
int i = 6;  
while(i >= 2) {  
    System.out.print(i+" ");  
    i--;  
}
```

A: 6 5 4 3 2 1 0

B: 6 5 4 3 2 1

C: 6 5 4 3 2

D: 6 5 4 3

E: 6 5

What does this print?

```
int i = 6;  
while(i >= 2) {  
    System.out.print(i+" ");  
    i--;  
}
```

A: 6 5 4 3 2 1 0

B: 6 5 4 3 2 1

C: 6 5 4 3 2

D: 6 5 4 3

E: 6 5

i	output
6	6
5	6 5
4	6 5 4
3	6 5 4 3
2	6 5 4 3 2
1	loop ends!

Done!

Extra slides after this

How many times does this while loop execute the print statement?

```
int i = 4;  
while(i >= 0) {  
    System.out.print("countdown " +i);  
    i++;  
}
```

A: 0

B: 1

C: 4

D: 6

E: the loop is infinite

How many times does this for loop execute the print statement?

```
int i = 4;  
while(i >= 0) {  
    System.out.print("countdown " +i);  
    i++;  
}
```

i keeps getting bigger!
i will always be >= 4

A: 0

B: 1

C: 4

D: 6

probably meant i--

E: the loop is infinite